

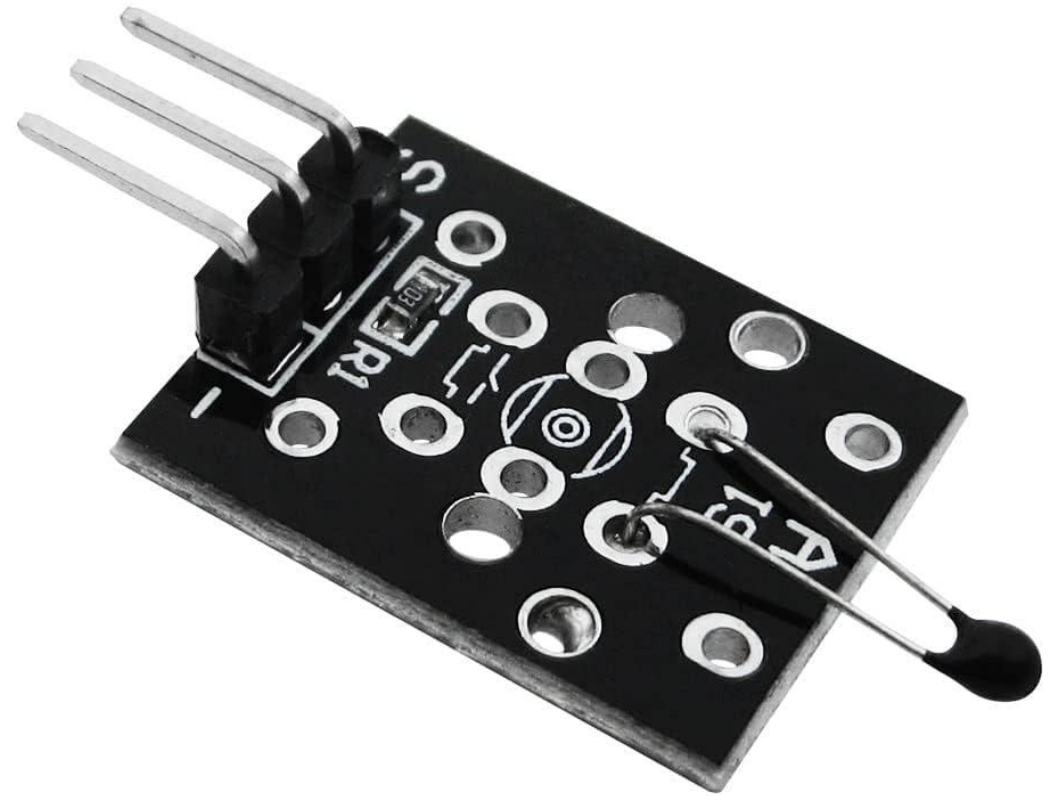
Internet of Things

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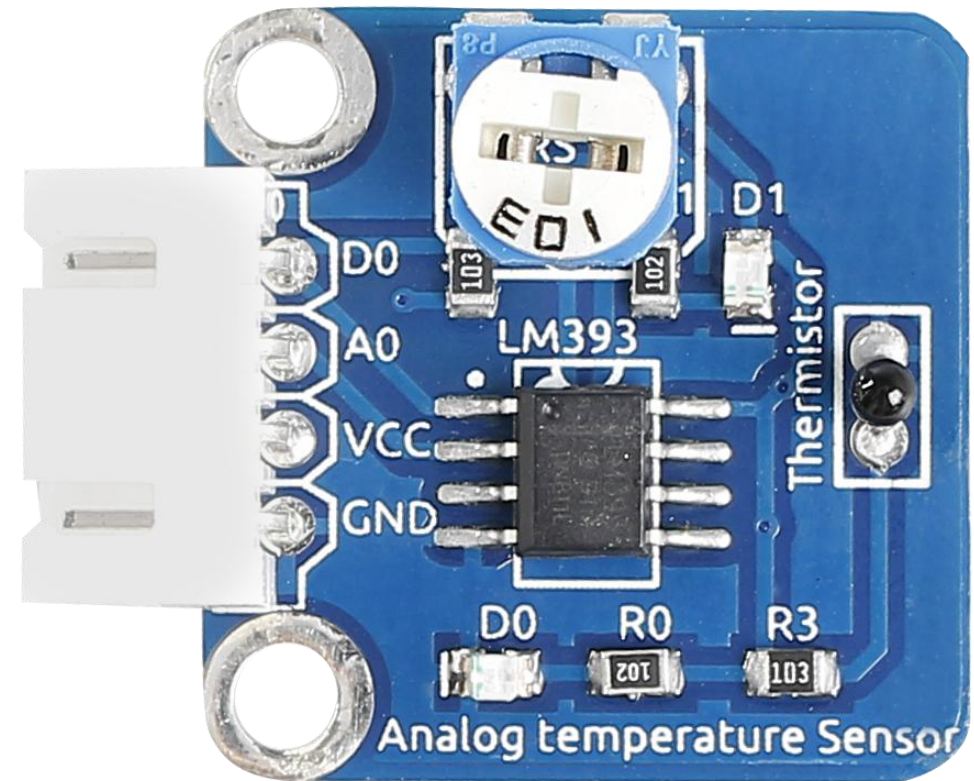
Temperature Sensor

- Analog Devices analog temperature sensors provide current or voltage output proportional to the absolute temperature with accuracies of up to $\pm 1^{\circ}\text{C}$.
- A temperature sensor is a component that senses temperature and converts it into output signals. By material and component features, temperature sensors can be divided into two types: thermal resistor and thermocouple.



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Thermistor is one kind of the former type. It is made of semiconductor materials; most thermistors are negative temperature coefficient (NTC) ones, the resistance of which decreases with rising temperature. Since their resistance changes acutely with temperature changes, thermistors are the most sensitive temperature sensors.



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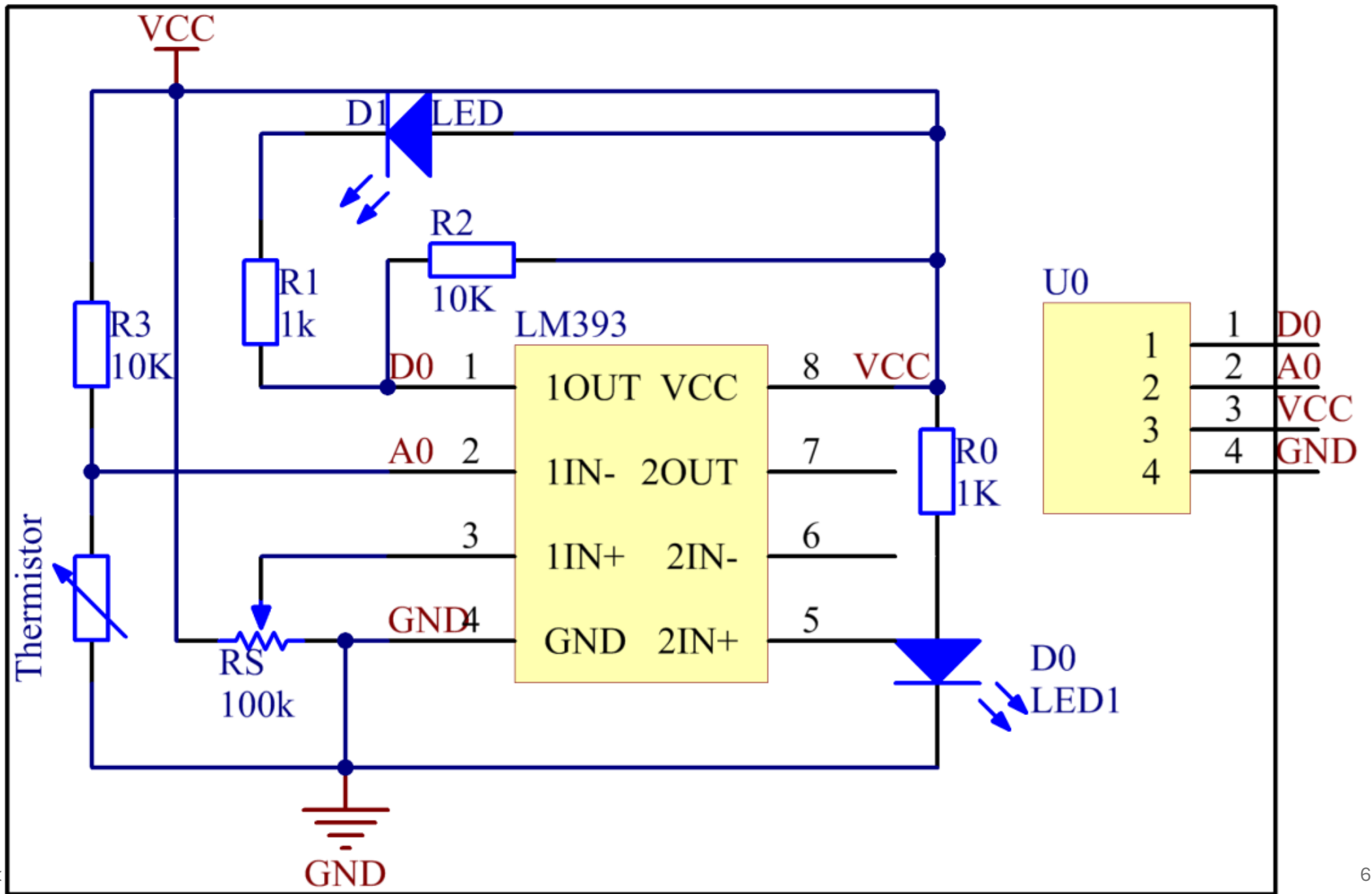
The Analog Temperature Sensor module uses an NTC thermistor, thus measuring temperature sensitively. It also has a built-in comparator LM393 which enables the module to output both digital and analog signals at the same time. The module can be used for temperature alarm and temperature measurement.

Working

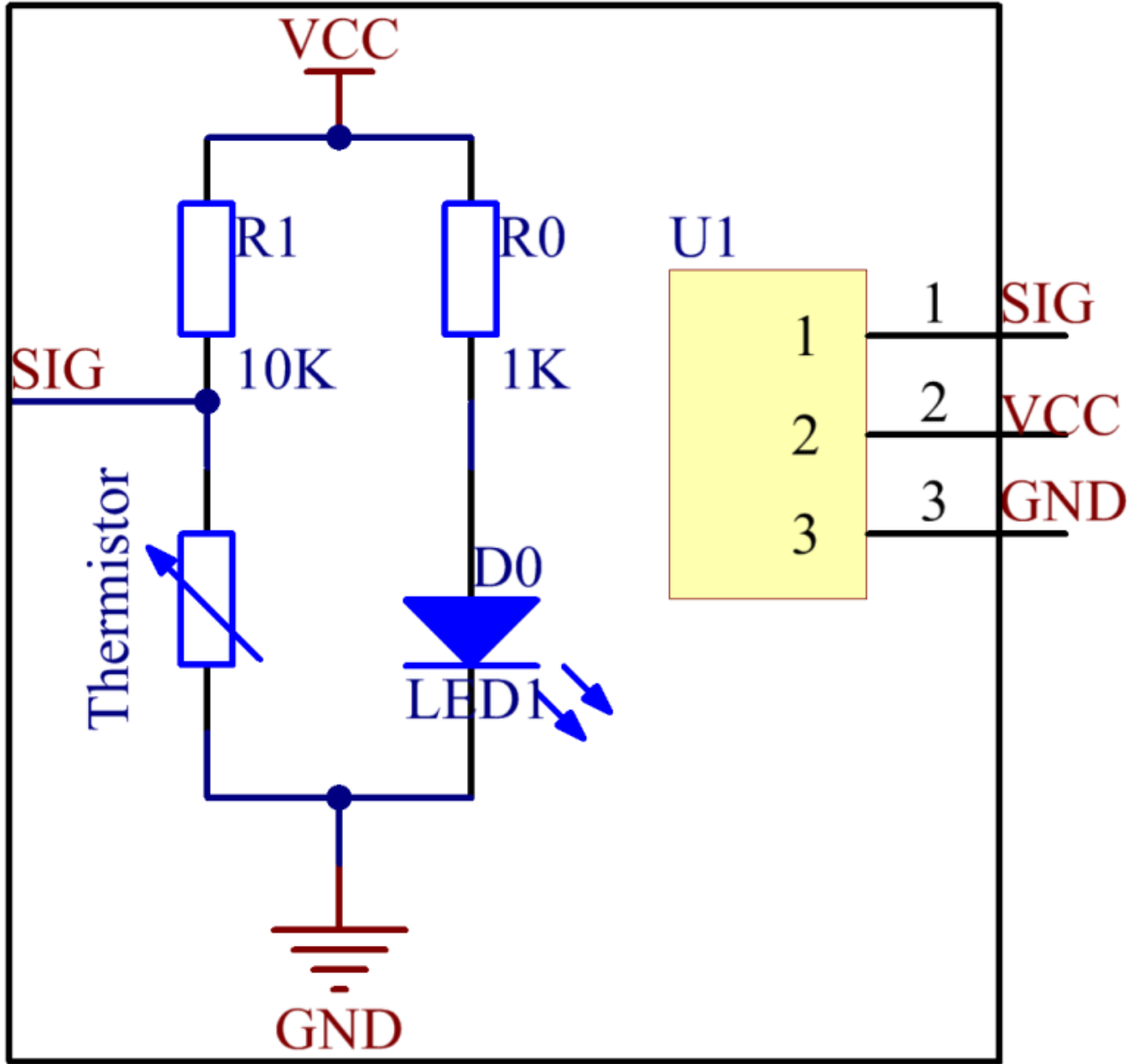
There is a comparator LM393 on the analog temperature sensor module. You may set a threshold by the potentiometer on it. When you touch the thermistor, the value of A0 decreases.

Once the value is lower than the threshold, D0 will output high level and the indicator LED on the analog temperature sensor module and that attached to pin 13 of the Uno will go out.

You can check the value of A0 and D0 on Serial Monitor.



Thermistor



To interface with the Temperature Sensor.

```
1.  const float BETA = 3950; // should match the Beta Coefficient of the thermistor
2.  void setup() {
3.      Serial.begin(9600);
4.  }
5.
6.  void loop() {
7.      int analogValue = analogRead(A0);
8.      float celsius = 1 / (log(1 / (1023. / analogValue - 1)) / BETA + 1.0 / 298.15) -
9.          273.15;
10.     Serial.print("Temperature: ");
11.     Serial.print(celsius);
12.     Serial.println(" °C");
13.     delay(1000);
14. }
```

Soil Moisture Sensor

- The soil moisture sensor is used to gauge the volumetric content of water within the soil.
- As the straight gravimetric dimension of soil moisture needs eliminating, drying, as well as sample weighting.
- These sensors measure the volumetric water content not directly with the help of some other rules of soil like dielectric constant, electrical resistance, otherwise interaction with neutrons, and replacement of the moisture content.
- The relation among the calculated property as well as moisture of soil should be adjusted & may change based on ecological factors like temperature, type of soil, otherwise electric conductivity.
- The microwave emission which is reflected can be influenced by the moisture of soil as well as mainly used in agriculture and remote sensing within hydrology.

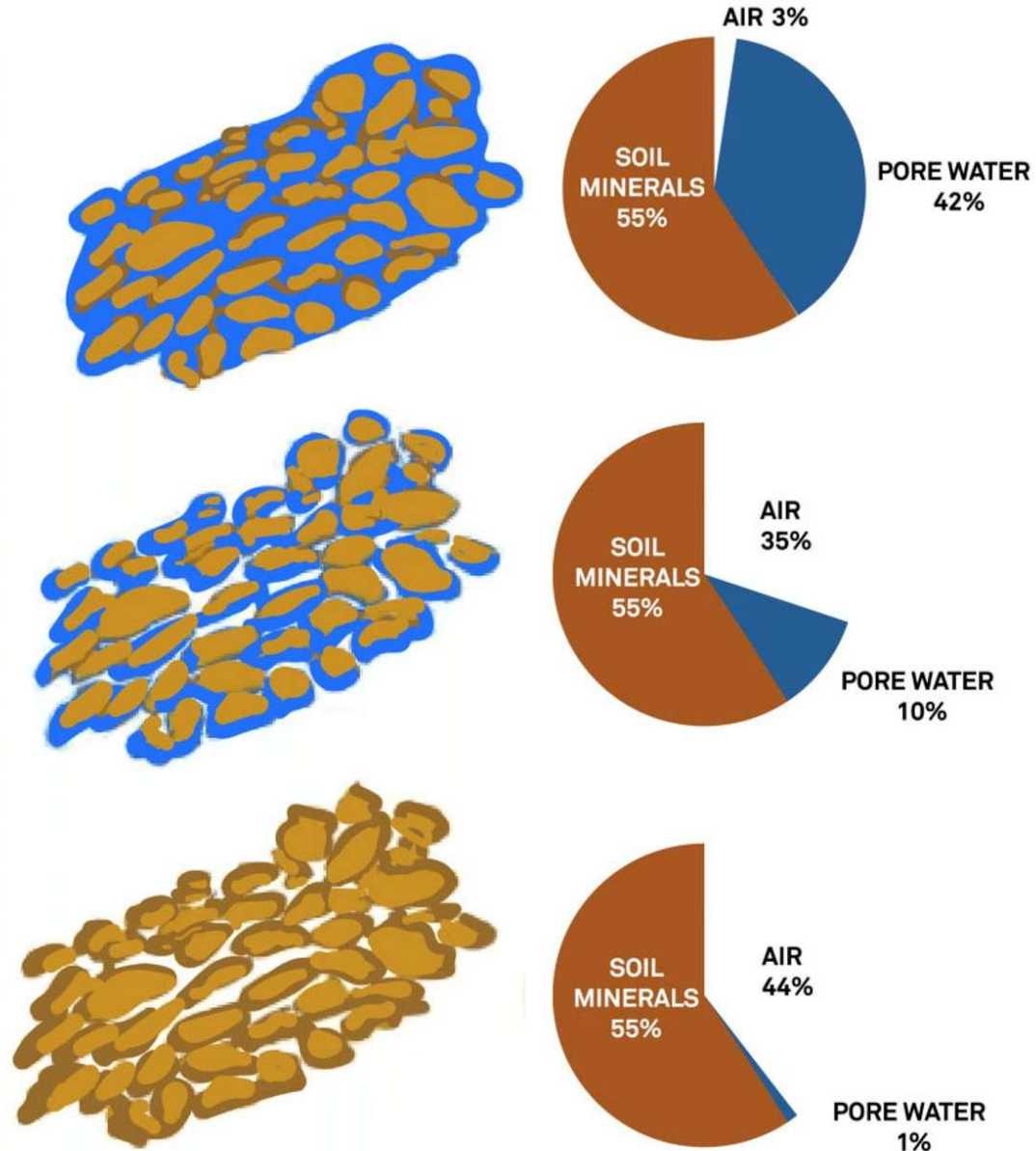
WATER CONTENT: AMOUNT

θ is volumetric water content (VWC)

V_w is the volume of water

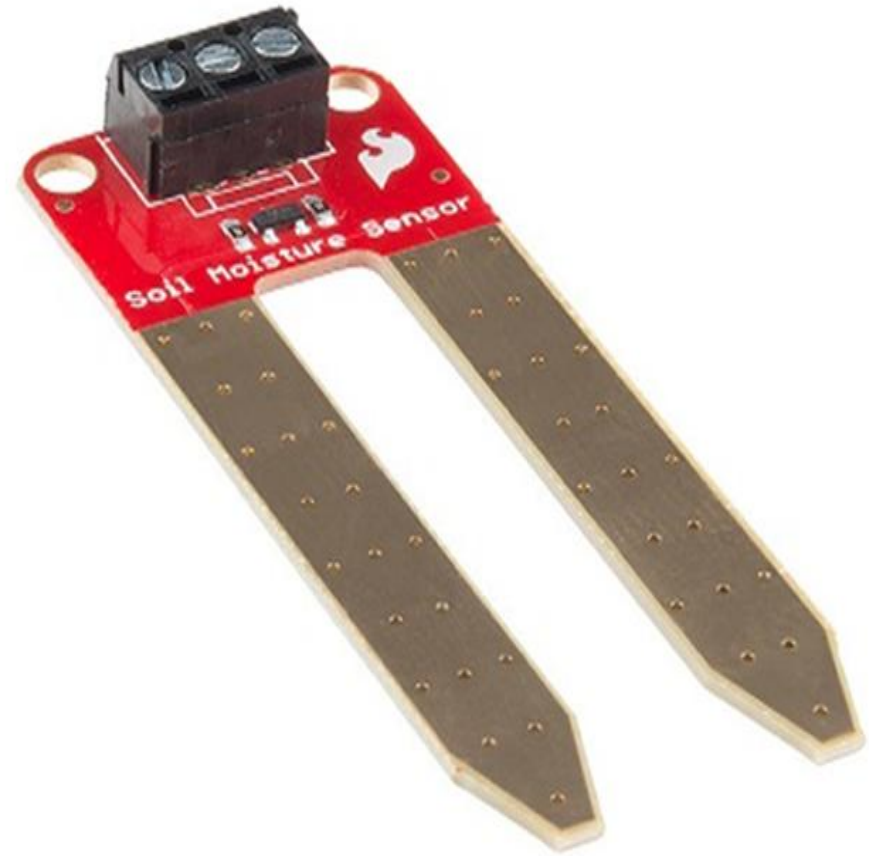
V_T is total sample volume

$$\theta = \frac{V_w}{V_T}$$



Contd.

These sensors normally used to check volumetric water content, and another group of sensors calculates a new property of moisture within soils named water potential. Generally, these sensors are named as soil water potential sensors which include gypsum blocks and tensiometer.



More About workings and reliability of the sensor: <https://metergroup.com/measurement-insights/soil-moisture-sensors-how-they-work-why-some-are-not-research-grade/>

Pin Configuration of Soil Moisture

- VCC pin is used for power
- A0 pin is an analog output
- D0 pin is a digital output
- GND pin is a Ground

This module also includes a potentiometer that will fix the threshold value, & the value can be evaluated by the comparator-LM393. The LED will turn on/off based on the threshold value.

Specification

The specification of this sensor includes the following.

- The required voltage for working is 5V
- The required current for working is <20mA
- Type of interface is analog
- The required working temperature of this sensor is 10°C~30°C

Working Principle of Soil Moisture Sensor

The moisture of the soil plays an essential role in the irrigation field as well as in gardens for plants. As nutrients in the soil provide the food to the plants for their growth. Supplying water to the plants is also essential to change the temperature of the plants. The temperature of the plant can be changed with water using the method like transpiration. And plant root systems are also developed better when rising within moist soil. Extreme soil moisture levels can guide to anaerobic situations that can encourage the plant's growth as well as soil pathogens. This article discusses an overview of the soil moisture sensor, working and its applications.

Assignment

- WAP to interface with Soil Moisture Sensor.
- WAP to interface using Analog and Digital part of the Sensor.
- WAP to Soil Moisture Sensor and Create 5 Diff ranges.
- WAP to Create 5 Different colors for the 5 Ranges.

Thanks